Auditing a Four-Class Chest X-ray Classifier for Shortcut Learning

Blessing Paul | November 2026 | Code: github.com/paulblessingjeremiah-ai/chest-xray-cnn-gradcam

Summary

A ResNet18 fine-tuned on a four-class chest X-ray set (COVID-19, normal, pneumonia, TB) reached 98.2% mean test accuracy over three seeds. Before trusting that number, I checked what the model could be using besides the lungs. The four classes come from different sources and differ in image size, markers, redaction boxes and patient age. A logistic regression that sees only image size, brightness and border statistics reaches 81% on the training images, with no anatomy at all. Masking everything outside the lungs lowers the classifier's accuracy to 94.2%. I conclude that part of the accuracy comes from non-anatomical cues. The result is not evidence that the model detects disease.

1. Data and model

The data has four classes: COVID-19, Normal, Pneumonia and TB, with 2,000 training images (500 per class) and 600 test images (150 per class). The collection appears to be compiled from several public sources. All 650 Pneumonia images are resized copies of images from the Kermany dataset, which contains only children aged one to five. The sources of the other three classes are unconfirmed.

The model is a ResNet18 pretrained on ImageNet, with its last layer replaced for four classes. It was trained for 10 epochs (Adam, learning rate 0.0001, batch size 32). I held out 15% of the training images as a validation set to choose the best epoch, and I used the test set once per run. Every result below is the mean of three random seeds. An earlier version of this project chose the best epoch on the test set and reported 99%. That number was optimistic, and this version replaces it.

2. The classes differ in how the images look

- Image size depends on the class: 299 x 299 pixels for COVID-19 and Pneumonia, and 512 x 512 for Normal and TB.

- In a random sample of eight images per class, the Pneumonia images look like children's X-rays with a large 'R' marker and timestamps. Six of the eight TB images have white rectangles covering part of the image. Several COVID-19 images carry text labels.
- I matched 32 x 32 thumbnails of every image against the 5,856 Kermany images. All 650 Pneumonia images matched a Kermany Pneumonia image (correlation 1.000). The best matches for COVID-19, Normal and TB were 0.94, 0.93 and 0.88, so they do not come from Kermany.
- The Pneumonia copies were also stretched to a square, and no other class has this distortion.

3. A baseline that sees no anatomy

I trained a logistic regression on seven numbers per image: height, width, aspect ratio, mean brightness, contrast, and the share of black and of white pixels. With five-fold cross-validation on the training images, chance is 25%.

Features	Accuracy
Size and shape only	50%
Brightness and borders only	54%
All seven	81%

Size alone gives exactly 50%. That is what you would expect if size separates the pair COVID-19 and Pneumonia from the pair Normal and TB, but not the classes inside each pair. The brightness and border measures separate most of the rest.

4. Duplicate images

Using the same thumbnails, I looked for near-copies (correlation above 0.98). Sixty-four training images duplicated other training images (COVID-19 52, Pneumonia 4, TB 8). Thirty-four test images were copies of training images or of each other (COVID-19 31, Pneumonia 1, TB 2). I removed them, which left a test set of 566 images. For the same models, accuracy was 98.22% on the full test set and 98.17% after removal. The duplicates did not inflate the result. Thumbnails can miss copies that were cropped or flipped, so this check has limits.

5. Masking everything outside the lungs

I used my own lung segmentation U-Net (test Dice 0.96, trained on 704 chest X-rays from the Montgomery and Shenzhen sets) to black out everything except the lungs. This removes borders, markers, timestamps, white boxes and the background. I retrained the classifier with the same settings on the de-duplicated data.

Input	Test accuracy (mean of 3 seeds)	Range	COVID-19 recall
Original images	98.2%	97.9% to 98.4%	94.1%
Lung region only	94.2%	94.2% to 94.4%	91.3%

The gap is about 3.9 points, and the ranges do not overlap. So part of the accuracy depends on information outside the lungs. This is an upper estimate of that effect. The U-Net was trained on adults and cut away some real lung tissue, especially in the Pneumonia images, where several masks were fragmented. Ragged mask shapes can themselves be a cue.

Accuracy stays well above chance with only the lungs visible, so there is also signal inside them. But the lungs still differ between classes in size and framing (TB lungs fill 44% of the image on average, against 30% to 33% for the others), in image processing, and in patient age.

6. What Grad-CAM adds

Grad-CAM maps are coarse, about 7 x 7 cells stretched over the image. For the COVID-19, Normal and TB examples, the strongest activation falls mostly inside the chest. For the Pneumonia example it sits in the lower corners and along the sides. In one earlier run, 13 of 600 test images were wrong, and all six I inspected involved COVID-19. One TB image was a small X-ray on a mostly blank canvas. For the same X-ray, two training runs with similar accuracy highlighted different lungs. Grad-CAM shows where a model looked. It cannot settle whether the model used a shortcut.

7. What the results show and do not show

They show that the classifier depends partly on information outside the lungs, that the four classes can be largely told apart without any anatomy (81%), and that duplicate images were not the cause. They do not show that the model detects disease, how it

would perform on X-rays from a new hospital or scanner, or which single cue it relies on.

8. What I would do next

- Train a model on the region outside the lungs only. High accuracy there would measure the shortcut directly.
- Build the classes from one source and one scanner type, or match them on age and image processing.
- Test on images from another hospital.
- Repeat the checks with more seeds, and use cross-validation across patients where patient identifiers exist.

References

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